YouTube Trending Video Analytics

Introduction

The project **YouTube Trending Video Analytics** aims to uncover patterns and trends in YouTube videos across multiple regions. By leveraging data-driven methods, this analysis provides actionable insights into viewer preferences, sentiment distribution, and the dynamics of video virality on YouTube’s trending list.

Abstract

This project consolidates YouTube trending datasets from various countries and analyzes them to identify trends, category performance, and viewer engagement levels. The process includes data cleaning, sentiment analysis on video titles and tags, SQL-based category ranking, and visualization through Tableau dashboards. The ultimate goal is to understand how content, sentiment, and region impact a video’s trending potential.

Tools Used

- Python (Pandas, Matplotlib, Seaborn) for data preprocessing and visualization
- SQL for data querying and ranking video categories by average views
- Tableau for creating interactive region-wise comparison dashboards
- VADER (NLTK) for performing sentiment analysis on video titles and tags

Steps Involved in Building the Project

1. Data Collection: Combine regional trending datasets (e.g., US, India, Canada).
2. Data Cleaning: Handle missing values, normalize columns, and merge datasets.
3. Sentiment Analysis: Use VADER to compute sentiment polarity scores for titles and tags.
4. SQL Analysis: Rank video categories based on average view count and engagement.
5. Visualization: Generate time-series plots of trending duration using Matplotlib and Seaborn.
6. Dashboard Creation: Build a Tableau dashboard showcasing popular genres, regional insights, and sentiment summaries.

Conclusion

The YouTube Trending Video Analytics project successfully demonstrates how data integration, sentiment analysis, and visualization can provide comprehensive insights into audience behavior across regions. The study concludes that sentiment-rich and engaging content significantly influences video virality, while regional preferences shape category popularity. These insights can help content creators optimize their strategy for better reach and engagement.